

Modern Methods in Video Anomaly Detection: A Review

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Introduction to Video Anomaly Detection

- **Video Anomaly Detection (VAD)** focuses on identifying events that deviate from normal patterns in surveillance, traffic, and safety-critical environments.
- VAD is challenging because anomalies are **rare**, **diverse**, and often **ill-defined**, requiring reliable and interpretable systems.
- **Existing deep learning-based VAD approaches** include supervised, weakly supervised, and unsupervised paradigms.
 - These methods often struggle with **performance** and **requirement of substantial data**.
- The rise of **Multimodal Large Language Models (MLLMs)** introduces new opportunities for VAD because of their ability to **jointly process and reason over visual and textual information**, opening promising new research directions.

-  **VERA: Explainable Video Anomaly Detection** (2024)
-  **HiProbe-VAD: A Tuning-Free MLLM-Based Framework for Video Anomaly Detection** (2025)
-  **EventVAD: Event-based Video Anomaly Detection** (2024)

VERA: Core Idea

Key Observation

Despite the promise of MLLMs, existing VLM-based VAD approaches face two main limitations:

- use of **external LLM reasoning modules** → high inference cost,
- reliance on **instruction tuning** → requires expensive frame-level annotations.

We need a method that keeps VLMs frozen while still enabling strong reasoning and interpretability.

Proposed Framework: VERA

VERA is introduced to overcome these challenges by enabling effective reasoning **without modifying or tuning the VLM itself**.

Key insight

VERA learns a collection of **guiding questions** that act as **trainable parameters**..

VERA: Core Idea (Cont.)

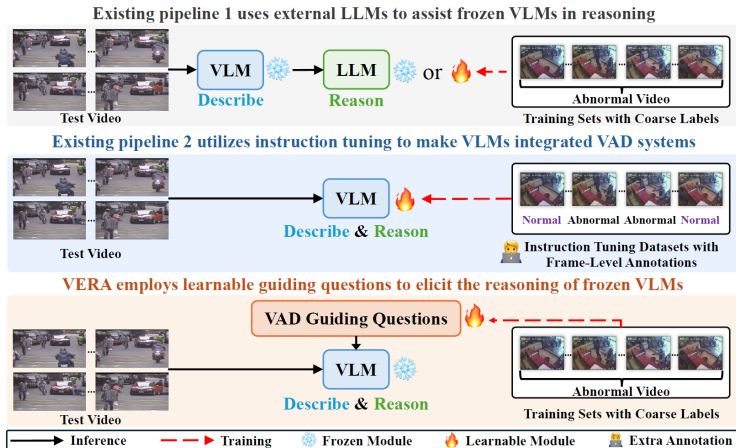


Figure: VERA renders frozen VLMs to describe and reason with learnable guiding questions learned from coarsely labeled data.

VAD: Problem Formulation

Video Anomaly Detection Setting

Let $V = \{I_i\}_{i=1}^F$ be a video containing F frames, where I_i is the i -th frame. The goal of VAD is to detect the start and end of anomalous events within the video.

Frame-Level Labels

- Each frame is assigned a binary label:

$$y_i \in \{0, 1\}$$

- $y_i = 1$ indicates an anomalous frame
- $y_i = 0$ indicates a normal frame

VERA Objective

VERA uses a frozen VLM f_{VLM} to predict continuous anomaly scores:

$$\hat{Y} = [\hat{y}_1, \dots, \hat{y}_F], \quad \hat{y}_i \in [0, 1]$$

VAD: Training Data

Coarsely Labeled Training Sets

VAD datasets typically provide only **video-level labels** rather than frame-level annotations.

- Training set:

$$D = \{(V^{(j)}, Y^{(j)})\}_{j=1}^N$$

where N is the number of training videos.

- For each video $V^{(j)}$:

$$V^{(j)} = \{I_i^{(j)}\}_{i=1}^{F_j}$$

- Label meaning:

- $Y^{(j)} = 1 \rightarrow$ video contains **at least one anomaly**
- $Y^{(j)} = 0 \rightarrow$ video contains **no anomalies**

- No fine-grained annotations such as frame-level y_i .

Training Objective

Goal

We aim to learn guiding questions that break down the complex and ambiguous concept of “**anomaly**” into identifiable anomalous patterns. These patterns vary across datasets, making manually designed descriptions ineffective for generalization.

- VERA proposes a general VL framework to automatically generate such guiding questions.
- Guiding question set:

$$Q = \{q_1, \dots, q_m\}, \quad q_i \text{ is the } i\text{-th question, } 1 \leq i \leq m$$

- The set Q is treated as **learnable parameters**.
- Optimization occurs through **verbal interaction** between:
 - a learner
 - an optimizer

both modeled by VLMs via instruction-following prompts.

Training Data

Training Data

Paired sampled frames and video-level labels are used. Sampling is required due to the large number of frames.

- Uniform sampling yields the best performance.
- For any video $V^{(j)} \in D$:

$$\ell = \left\lfloor \frac{F_j}{S} \right\rfloor$$

where S is the number of sampled frames.

- Sampled frame sequence:

$$\tilde{V}^{(j)} = [I_1^{(j)}, I_{\ell+1}^{(j)}, \dots, I_{(S-1)\ell+1}^{(j)}]$$

- Training pairs:

$$\{(\tilde{V}^{(j)}, Y^{(j)})\}_{j=1}^N$$

Updating Q via Learner and Optimizer

Verbalized Optimization (VML)

Since guiding questions Q are **verbal expressions of anomaly patterns**, VERA optimizes them via verbal communication between:

$$f_{\text{learner}} \quad \text{and} \quad f_{\text{opt}}$$

instead of numerical optimizers (e.g., Adam).

General VL Formulation

Any LLM-based model is written as:

$$f(x; \phi)$$

where:

- x : input data
- ϕ : natural-language instructions (treated as learnable parameters)

Learner Agent

The learner is modeled by the frozen VLM f_{VLM} with prompt template θ :

$$f_{\text{learner}}^{(t)}(x) = f_{\text{VLM}}(x; (\theta, Q_t))$$

- x : input for the learning task
- (θ, Q_t) : task prompt + current guiding questions
- Q_t : guiding questions at iteration t

Optimizer Agent

The optimizer evaluates predictions and refines Q_t :

$$f_{\text{opt}}^{(t)}(z) = f_{\text{VLM}}(z; (\psi, Q_t))$$

- z : optimizer input
- ψ : instruction to improve Q_t
- $f_{\text{learner}}^{(t)} \neq f_{\text{opt}}^{(t)}$

Forward Pass of the Learner

The learner performs the forward pass and outputs a prediction using only the **coarsely labeled training data**.

- A **binary classification task** is designed for the learner.
- It considers:
 - temporal nature of video data
 - sparsity of anomalies
 - weak supervision in VAD datasets
- The task description is written in the **Model Description** section of the prompt template θ .
- The guiding questions Q_t are inserted into the **Prompt Questions** section to trigger reasoning in the VLM.

Backward Pass of the Optimizer

The optimizer performs the **backward pass** to update the guiding questions Q_t using a mini-batch of size n .

- Let the batched visual inputs be:

$$V_{\text{batch}} = [\tilde{V}_{\text{batch}}^{(1)}, \dots, \tilde{V}_{\text{batch}}^{(n)}]$$

- The corresponding ground-truth video-level labels:

$$Y_{\text{batch}} = [Y_{\text{batch}}^{(1)}, \dots, Y_{\text{batch}}^{(n)}]$$

Optimization Step in f_{opt} (Cont.)

Learner Predictions

Using current Q_t , the learner produces:

$$\hat{Y}_{\text{batch}} = [\hat{Y}_{\text{batch}}^{(1)}, \dots, \hat{Y}_{\text{batch}}^{(n)}]$$

Optimizer Update Rule

After reading the batched data and predictions, the optimizer generates a new question set:

$$Q_{t+1} = f_{\text{opt}}^{(t)}(V_{\text{batch}}, \hat{Y}_{\text{batch}}, Y_{\text{batch}})$$

where Q_{t+1} is produced through the text-generation and instruction-following abilities of the frozen VLM.

Overall Training Pipeline

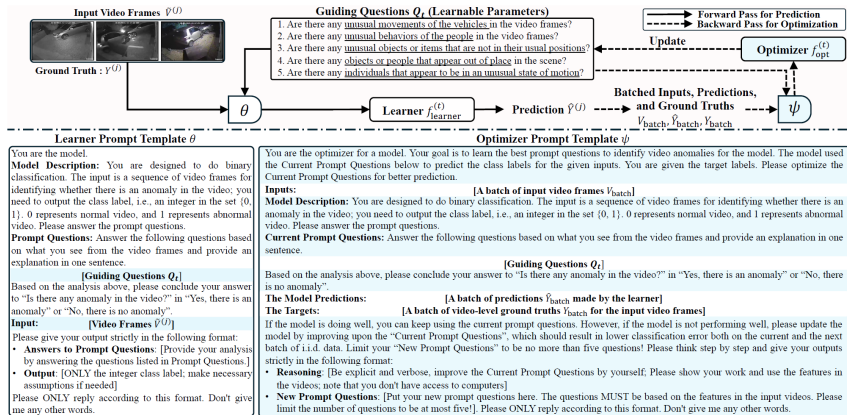
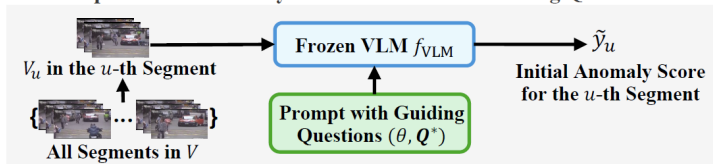


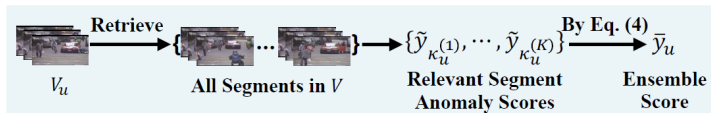
Figure: The overall training pipeline in VERA aims to optimize VAD guiding questions iteratively. In each iteration t , the optimization is verbalized by providing verbal instructions for the learner and optimizer to follow. They will generate predictions and new guiding questions, respectively.

Inference in VERA

Step 1: Initial Anomaly Scores via Learned Guiding Questions



Step 2: Ensemble Segment-Level Anomaly Scores with Scene Context



Step 3: Frame-Level Anomaly Scoring with Temporal Context

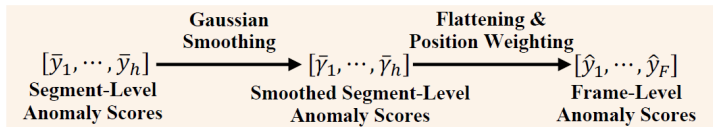


Figure: VERA computes anomaly scores with Q in three steps.

Inference in VERA: Step 1

Initial Anomaly Scores via Learned Guiding Questions

VERA divides the video into segments and analyzes each segment independently to compute initial anomaly scores.

- **Segmentation:** Equidistant frame sampling is performed within the video to obtain a set of segment centers.
- **Sampling:** For each center frame, a 10-second window is defined as a segment, from which 8 frames are uniformly sampled (V_u).
- **Initial Scoring:** The sampled frames are passed to the frozen VLM alongside the optimized guiding questions (Q^*) and prompt template (θ).
- **Prediction:** The initial anomaly score for the u -th segment is computed as:

$$\tilde{y}_u = f_{\text{VLM}}(V_u; (\theta, Q^*))$$

where $\tilde{y}_u \in \{0, 1\}$ indicates whether the VLM identifies an anomaly in the segment.

Inference in VERA: Step 2

Ensemble Segment-Level Scores with Scene Context

Because initial scores only examine short, isolated moments, VERA refines them by incorporating broader scene context (e.g., similar backgrounds or actors)].

- **Relevance Measurement:** The cosine similarity of visual feature representations (extracted via a pretrained extractor like ImageBind) is used to find the top- K relevant segments.
- **Refinement Mechanism:** The initial score of the current segment is updated into an ensemble score by weighting the initial scores of the retrieved segments.

Ensemble Equation

Using the indices of the top- K segments (κ_u), the refined score $\bar{y}_u$ is:

$$\bar{y}_u = \sum_{i=1}^K \tilde{y}_{\kappa_u^{(i)}} \cdot \frac{\exp(\text{sim}(u, \kappa_u^{(i)})/\tau)}{\sum_{j=1}^K \exp(\text{sim}(u, \kappa_u^{(j)})/\tau)}$$

where the weights are normalized via a Softmax function with temperature τ .

Frame-Level Anomaly Scoring with Temporal Context

To accurately capture how abnormal events evolve over time, VERA integrates temporal context to compute fine-grained frame-level scores.

- **Local Context (Gaussian Smoothing):** Segment-level scores $\bar{Y}$ are aggregated using a Gaussian kernel G , yielding smoothed scores $\bar{\Gamma} = \bar{Y} * G$.
- **Flattening:** These smoothed segment scores are then assigned to each individual frame within the segment to create a flattened frame-level sequence $[\rho_1, \dots, \rho_F]$.
- **Global Context (Position Weighting):** A Gaussian function $w(i)$ is applied to diminish scores near the beginning and end of the event, reflecting the anomaly's natural progression.
- **Final Frame-Level Score:** The final anomaly score for the i -th frame is calculated as:

$$\hat{y}_i = w(i) \cdot \rho_i$$

yielding the final continuous score sequence $\hat{Y} = [\hat{y}_1, \dots, \hat{y}_F]$.

Implementation Details

VERA Configuration

- Backbone VLM: **InternVL2-8B** by default
- Additional backbones for ablation: **Qwen2-VL-7B**, **InternVL2-40B**
- Number of guiding questions: $m = 5$
- Batch size: $n = 2$
- Sampled frames per video: $S = 8$

Training Setup

- VERA trains guiding questions using coarsely labeled videos.
- Training is performed for no more than **10 epochs**.
- Validation accuracy is checked every **100 iterations**.
- The best guiding question set Q^* is selected based on validation performance.

Comparison to State-of-the-art Methods on UCF-Crime

Method	AUC (%)
<i>Non-explainable VAD Methods</i>	
MGFN	86.98
SSRL	87.43
CLIP-TSA	87.58
<i>Explainable VAD Methods</i>	
VERA	86.55
LAVAD	80.28
LLAVA-1.5	72.84

Table: AUC (%) on UCF-Crime.

Comparison to State-of-the-art Methods on XD-Violence

Method	AUC (%)
<i>Non-explainable VAD Methods</i>	
BODS	57.32
GODS	61.56
RareAnom	68.33
<i>Explainable VAD Methods</i>	
VERA	88.26
LAVAD	85.36
LLAVA-1.5	79.62

Table: AUC (%) on XD-Violence.

Ablation Study: Sampling Strategies

Strategy	AUC (%)
Random	83.63
TSN	82.63
Uniform	86.55

Table: Sampling strategies explored in VERA training.

Observation

Uniform sampling performs the best because it preserves temporal structure and maintains consistent motion patterns throughout the long video.

Ablation Study: Guiding Questions

Question Type	AUC (%)
No questions	78.81
Manually written questions by human	81.15
Learned questions w/o iteratively inputting V_{batch}	78.06
Iteratively learned questions (used in VERA)	86.55

Table: The way we obtain guiding questions affects AUC substantially.

Observation

Iteratively learned guiding questions are critical for performance. Removing them or learning them without iterative optimization causes a large drop in AUC.

Ablation Study: Effect of the Number of Guiding Questions

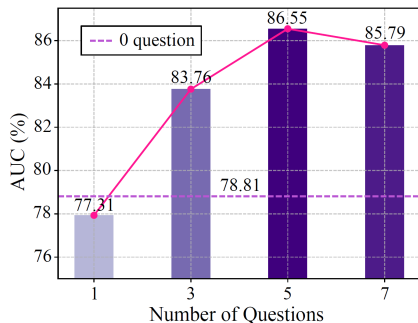


Figure: Effect of the number of guiding questions on AUC.

Observation

Performance improves as the number of guiding questions increases up to $m = 5$, and then saturates.

Ablation Study: Inference Pipeline

Operation	AUC (%)
Initial (Step 1)	76.10
Initial + Retrieval (Step 2)	84.53 (+8.43)
Initial + Retrieval + Smoothing (Step 3)	85.48 (+0.95)
Initial + Retrieval + Smoothing + Weighting (Step 3)	86.55 (+1.07)

Table: Ablation study of each step in VERA's inference.

HiProbe-VAD: Core Idea

Key Observation

Existing MLLM-based VAD methods face critical limitations:

- reliance on **output-layer features** → weak anomaly discrimination,
- dependence on **model fine-tuning** → costly, dataset-specific, and not scalable.

There is a need for a tuning-free approach that leverages MLLMs effectively while preserving generalization.

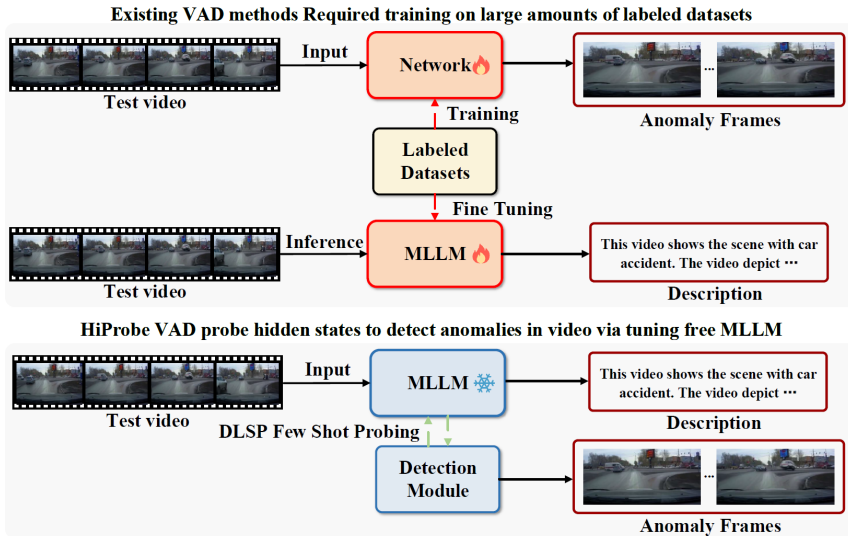
Proposed Framework: HiProbe-VAD

HiProbe-VAD overcomes these limitations by probing **intermediate hidden states** of pre-trained MLLMs—achieving strong anomaly detection **without any model tuning**.

Key Insight

HiProbe leverages the **Intermediate Layer Information-Rich Phenomenon**.

HiProbe-VAD: Core Idea (Cont.)



Intermediate Layer Information-Rich Phenomenon

Definition

Recent studies on Large Language Models (LLMs) show that **intermediate hidden layers often encode richer and more informative representations** than the final output layers.

Why It Happens

Intermediate layers provide a better trade-off between:

- **information retention**
- **noise reduction**

This emerges through mechanisms such as **compression** and **feature distillation**, making these layers more useful for downstream understanding.

Information-Rich Phenomenon for VAD

- Motivated by the richness observed in intermediate layers of LLMs, this work investigates whether a similar **information-rich phenomenon** exists in **Multimodal Large Language Models (MLLMs)** for video anomaly detection (VAD).
- **Hypothesis:** Intermediate layers in MLLMs may yield more nuanced representations of anomalies than final output layers, which are often optimized for text generation and may overlook subtle anomaly cues
- **Exploration Strategy:**
 - Systematically extract hidden state representations from each layer of pre-trained MLLMs (e.g., InternVL2.5) for benchmark VAD videos (XD-Violence, UCF-Crime).
 - For each input video, a single forward pass is performed to obtain hidden states h_l at each layer l .
 - The quality and utility of these representations are evaluated for distinguishing normal vs. anomalous videos using statistical and geometric analyses.

Anomaly Sensitivity via KL Divergence

Idea: Measure the **distributional difference** between normal and anomalous hidden states.

For feature dimension d at layer l , assuming Gaussian distributions:

$$h_{l,d}^N \sim \mathcal{N}(\mu_{l,d}^N, (\sigma_{l,d}^N)^2), \quad h_{l,d}^A \sim \mathcal{N}(\mu_{l,d}^A, (\sigma_{l,d}^A)^2)$$

KL Divergence (per dimension):

$$D_{\text{KL}}^{(d)}(l) = \frac{1}{2} \left[\log \frac{(\sigma_{l,d}^A)^2}{(\sigma_{l,d}^N)^2} + \frac{(\sigma_{l,d}^N)^2 + (\mu_{l,d}^N - \mu_{l,d}^A)^2}{(\sigma_{l,d}^A)^2} - 1 \right]$$

Overall KL at layer l :

$$D_{\text{KL}}(l) = \frac{1}{D} \sum_{d=1}^D D_{\text{KL}}^{(d)}(l)$$

Interpretation: A higher $D_{\text{KL}}(l)$ indicates a **greater statistical separation** between normal and anomalous features at layer l .

Class Separability via Local Discriminant Ratio

Idea: Evaluate how well hidden states can **linearly separate** normal and anomalous samples.

LDR (per feature dimension):

$$\text{LDR}^{(d)}(l) = \frac{(\mu_{l,d}^N - \mu_{l,d}^A)^2}{(\sigma_{l,d}^N)^2 + (\sigma_{l,d}^A)^2 + \epsilon}$$

Overall LDR at layer l :

$$\text{LDR}(l) = \frac{1}{D} \sum_{d=1}^D \text{LDR}^{(d)}(l)$$

Interpretation: A higher $\text{LDR}(l)$ implies **stronger linear separability** between normal and anomalous classes, indicating more discriminative representations.

Information Concentration via Feature Entropy

Idea: Measure how **diverse and information-rich** the hidden representations are.

For each feature dimension d at layer l , feature values are discretized into B bins.

Entropy (per dimension):

$$H^{(d)}(l) = - \sum_{j=1}^B p(h_l[d] \in \text{bin}_j) \log_2 p(h_l[d] \in \text{bin}_j)$$

Overall Entropy at layer l :

$$H(l) = \frac{1}{D} \sum_{d=1}^D H^{(d)}(l)$$

Interpretation: Higher entropy indicates a **more uniform and diverse distribution** of feature values, reflecting richer information content for anomaly detection.

Statistical Analysis on the Datasets

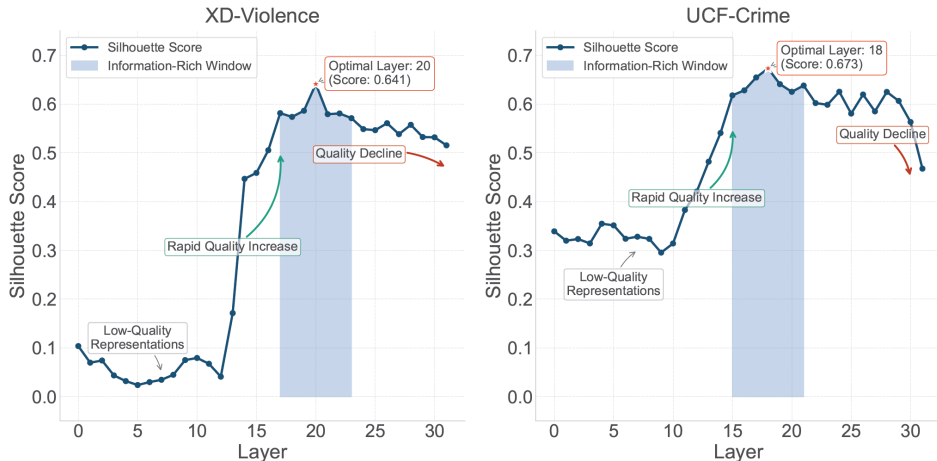


Figure: Silhouette score across layers on XD-Violence and UCF-Crime datasets, revealing a trend of increasing linear separability, peaking at intermediate layer 20 before declining in deeper layers.

HiProbe-VAD: Tuning-Free Video Anomaly Detection

HiProbe-VAD is a **tuning-free** framework that leverages the **information-rich intermediate layers** of pre-trained MLLMs for video anomaly detection.

Core Components:

- **Dynamic Layer Saliency Probing (DLSP)** to identify the most informative intermediate layer.
- **Lightweight Anomaly Scorer** trained with few-shot probing to classify frames.
- **Temporal Anomaly Localization** to detect anomalous frame intervals.

Preparation Phase (Before Inference):

- Extract hidden states from multiple layers of the MLLM.
- Identify the **optimal layer** for anomaly detection.
- Train a lightweight scorer using only a **small subset** of training videos (1%).

Dynamic Layer Saliency Probing

The Dynamic Layer Saliency Probing (DLSP) module aims to identify the intermediate layer l^* that provides the most **discriminative features** for distinguishing normal and abnormal video content.

Procedure:

- Use only **about 1%** of videos from UCF-Crime and XD-Violence.
- For each video v in this subset, extract hidden states $H^{(v,l)}$ at every layer l during the first-token generation of the MLLM.
- For each layer, compute:
 - anomaly sensitivity via **KL divergence**,
 - class separability via **LDR**,
 - information concentration via **Entropy**,

DLSP evaluates all intermediate layers to determine which one best separates normal and abnormal patterns.

Dynamic Layer Saliency Probing (Cont.)

Z-score Normalization:

For each metric $M \in \{D_{\text{KL}}(I), \text{LDR}(I), H(I)\}$:

$$\text{Norm}(M(I)) = \frac{M(I) - \mu_M}{\sigma_M}$$

where μ_M and σ_M are the mean and standard deviation across layers $1, \dots, L$.

Saliency Score per Layer:

$$S(I) = \text{Norm}(D_{\text{KL}}(I)) + \text{Norm}(\text{LDR}(I)) + \text{Norm}(H(I))$$

Optimal Layer Selection:

$$I^* = \arg \max_{I \in \{1, \dots, L\}} S(I)$$

This video-level analysis selects a robust, generalizable layer used for training the anomaly scorer and for real-time inference.

Lightweight Anomaly Scorer Training

The anomaly scorer uses a lightweight **logistic regression** classifier trained offline on hidden states from the optimal layer l^* selected by DLSP.

Prediction: For the i -th sample with resampled hidden state $h_{l^*}^{(i)}$:

$$p_i = \sigma \left(w^T h_{l^*}^{(i)} + b \right)$$

where $\sigma(\cdot)$ is the sigmoid function.

Training Objective: Binary cross-entropy loss over N samples:

$$L(w, b) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

The classifier is trained using **LBFGS** for 1000 epochs, producing a simple but effective model that leverages anomaly-sensitive features from the selected MLLM layer while remaining efficient for real-time inference.

Frame-Level Anomaly Scoring

For an input video, we segment it into consecutive parts and uniformly sample keyframes from each segment. For each sampled keyframe F_i , a single forward pass through the MLLM extracts hidden states $h_{l^*}(F_i)$ from the optimal layer l^* .

These hidden states are fed into the Lightweight Anomaly Scorer to compute the anomaly probability for each segment:

$$A(F_i) = \sigma(w^T h_{l^*}(F_i) + b)$$

where σ is the sigmoid function and (w, b) are the learned parameters of the logistic regression model.

This produces a temporal sequence of frame-level anomaly probabilities, indicating the likelihood of each keyframe being **anomalous**.

Temporal Anomaly Localization

To localize anomalies, HiProbe-VAD aggregates frame-level anomaly scores over time.

- First, apply **Gaussian smoothing** to reduce noise and obtain a smooth anomaly curve $C(t)$.
- Next, determine an adaptive threshold T based on the mean μ_A and standard deviation σ_A of anomaly scores derived from the DLSP few-shot training data:

$$T = \mu_A + \kappa \cdot \sigma_A$$

Frames with smoothed scores above T form **anomalous segments**, while those below form **normal segments**.

Explainable VAD via MLLMs

To provide human-interpretable insights, HiProbe-VAD performs auto-regressive reasoning using pre-trained MLLMs.

- Anomalous segments and normal segments are processed **separately**.
- Each segment is fed into the MLLM to generate natural- language descriptions.
- This produces **precise and context-aware explanations** detailing abnormal behaviors and contrasting them with normal activity.

This explanation step enhances the interpretability of the entire HiProbe-VAD pipeline and improves user understanding of the detected anomalies.

HiProbe-VAD: Overall Pipeline

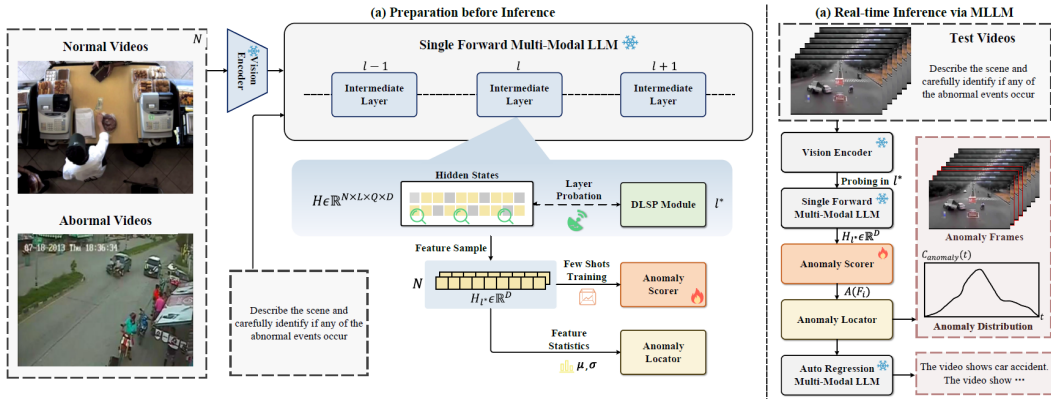


Figure: Overview of the HiProbe-VAD framework. HiProbe-VAD operates in two phases: (1) Offline preparation (DLSP and scorer training) and (2) Real-time inference (frame-level scoring and localization). The DLSP module assists to input layer-wise hidden states to the scorer, which then collaborates with the temporal locator to generate final detections and descriptions.

Implementation Details

- **Keyframe Sampling:** We uniformly sampled $K = 8$ keyframes at a fixed interval of each video segment of 24 frames.
- **Backbone MLLM:**
 - Primary: **InternVL2.5**.
 - Comparative: Qwen2.5-VL, LLaVA-OneVision, and Holmes-VAU.
- **Training Strategy:**
 - **Lightweight Scorer:** Logistic regression classifier trained on hidden states extracted from the optimal layer identified by DLSP.
 - **Solver:** Optimized using LBFGS for 1000 epochs.
- **Hyperparameters:**
 - Gaussian kernel width (σ): 0.4.
 - Threshold parameter (κ): 0.2.
 - Hardware: All experiments performed on an **NVIDIA 4090 GPU**.

Performance Comparison: UCF-Crime

Comparison on UCF-Crime dataset (AUC %)

Mode	Methods	AUC (%)
Weakly Supervised	Wu et al.	82.44
	RTFM	84.30
	S3R	85.99
	MSL	85.30
	VadCLIP	88.02
Tuning-Free Multimodal VAD	LAVA-1.5	72.84
	HiProbe-VAD (Qwen2.5-VL)	85.89
	HiProbe-VAD (InternVL2.5)	86.72
Fine-Tuned MLLM	HiProbe-VAD (Holmes-VAU)	88.91

Performance Comparison: XD-Violence

Comparison on XD-Violence dataset (AP %)

Mode	Methods	AP (%)
Weakly Supervised	RTFM	77.81
	MFGN	79.19
	S3R	80.26
	Yang et al.	83.68
	VadCLIP	84.51
Tuning-Free Multimodal VAD	LAVA-1.5[24]	50.26
	LAVAD[65]	62.01
	HiProbe-VAD (Qwen2.5-VL)	80.94
	HiProbe-VAD (InternVL2.5)	82.15
Fine-Tuned MLLM	HiProbe-VAD (Holmes-VAU)	89.51

Ablation Study on HiProbe-VAD

Ablation Setting	UCF-Crime AUC (%)	XD-Violence AP (%)
Full HiProbe-VAD	86.72	82.15
<i>w/o. Dynamic Layer Saliency Probing (DLSP)</i>		
Fixed Last Layer	83.21	79.28
Fixed Mid Layer (Layer 16)	78.60	75.61
<i>w/o. Lightweight Anomaly Scorer</i>		
SVM	84.87	80.63
Distance-based Scoring	80.34	75.65
<i>w/o. Temporal Localization</i>		
Fixed Threshold = 0.75	70.42	65.43
Fixed Threshold = 0.5	85.78	79.93
Fixed Threshold = 0.25	76.01	72.45

Impact of Sampling Rate on HiProbe-VAD

Sampling Rate (K)	AUC (%)
K = 2	76.20
K = 4	82.51
K = 8	86.72
K = 16	87.01

- Performance improves as the sampling rate increases.
- **K = 8** offers a strong trade-off between accuracy and efficiency.
- Although **K = 16** gives the highest AUC, it is more computationally expensive.

EventVAD: Core Idea

Key Observation

Existing MLLM-based methods (e.g., LAVAD) evaluate videos via frame-by-frame analysis. They face two main limitations:

- Image-based MLLMs struggle to model **temporal dynamics** and diverse event patterns, resulting in prediction inconsistency across frames.
- Reliance on a redundant multi-stage VQA pipeline, which is **inefficient** in both storage and inference.

Proposed Framework: EventVAD

EventVAD is a training-free framework that segments long videos into short events to enhance **event consistency** during MLLM understanding.

Key Insight

Combining tailored dynamic graph architectures and multimodal LLMs enables **fine-grained temporal-event reasoning**.

EventVAD: Core Idea (Cont.)

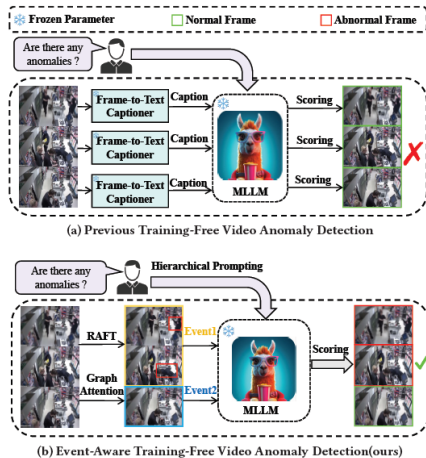


Figure: Existing methods struggle to localize fine-grained abnormal frames. EventVAD segments events to enhance event consistency and fine-grained abnormal detection.

EventVAD: Problem Formulation

Video Anomaly Detection Setting

Given an input video sequence $V = \{I_t\}_{t=1}^T$ with T frames, the objective is to detect anomaly frames. Anomaly indicators are denoted as $\{a_t\}_{t=1}^T$, where $a_t \in \{0, 1\}$ represents the binary label for frame I_t .

Overall Pipeline

1. **Event-Aware Dynamic Graph Construction:** Constructs a dynamic graph with multimodal temporal decay (nodes = fused features, edges = spatiotemporal connections).
2. **Graph Attention Propagation:** Refines node features through orthogonally constrained messaging.
3. **Statistical Boundary Detection:** Identifies event boundaries using unsupervised statistical features.
4. **Event-Centric Anomaly Scoring:** Uses semantic cues to assess anomalies within detected event units.

1) Event-Aware Dynamic Graph Construction

To address static graph limitations and slow training speeds, a **dynamic spatiotemporal graph model** $G = (F, E)$ is proposed.

- **Vertices (F):** Video frame units with enhanced feature encoding.
- **Edges (E):** Dynamically established using cross-modal similarity metrics and time-sensitive connections.

Node Feature Construction: Dual-Stream Architecture

1. Semantic Branch (CLIP): Amplifies feature discontinuities for detecting boundaries. Generates 512-dim L2-normalized embeddings:

$$f_i^{\text{clip}} = \frac{\phi_{\text{CLIP}}(I_i)}{\|\phi_{\text{CLIP}}(I_i)\|_2}$$

2. Motion Branch (RAFT): Captures temporal dynamics via optical flow, mapped to 128-dim vectors via projection matrix P :

$$f_i^{\text{flow}} = P^{\top} \mathbb{E}[o_{i \rightarrow i-1}]$$

1) Event-Aware Dynamic Graph Construction (Cont.)

Fused Node Representation

The final node representation for graph vertex $v_i \in F$ integrates semantic and motion constraints to define transformation boundaries (The semantic-motion fusion coefficient: $\alpha = 0.75$):

$$f_i = \alpha f_i^{\text{clip}} \oplus (1 - \alpha) f_i^{\text{flow}}$$

Temporal Decay Adjacency Matrix

To capture node relationships, the adjacency matrix E incorporates time decay to penalize remote associations:

$$E_{ij} = \frac{\alpha \cdot \cos(f_i^{\text{clip}}, f_j^{\text{clip}}) + (1 - \alpha) \cdot \exp(-\|f_i^{\text{flow}} - f_j^{\text{flow}}\|)}{1 + \gamma \cdot |i - j|}$$

- **Numerator:** Emphasizes short-term correlations via multimodal similarity.
- **Denominator:** Introduces a time decay factor γ to diminish the influence of distant frames.

2) Graph Attention Networks Propagation

Objective

To improve frame-by-frame representation while preserving temporal consistency **without** the need for additional model training.

Proposed Mechanism

A **training-free attention mechanism** based on orthogonal feature projection in spacetime.

- Builds upon **Graph Attention Networks (GAT)** with orthogonal constraints.
- Amplifies segment contrast through **graph-guided message passing**.
- Substantially enhances **event boundary distinguishability**.

Orthogonal Feature Projection

Preventing Dimension Collapse

Features from the previous step are projected into orthogonal subspaces using fixed matrices. This maximizes feature retention.

The orthogonal projection matrices are defined via QR decomposition:

$$\begin{cases} Q = \text{QR}(\mathcal{N}(0, 1)^{d \times k}) \\ K = \text{QR}(\mathcal{N}(0, 1)^{d \times k}) \\ V = \text{QR}(\mathcal{N}(0, 1)^{d \times d}) \end{cases}$$

- $\text{QR}(\cdot)$: Orthonormal columns obtained via QR decomposition.
- $d = 640$: Fused feature dimension.
- $k = 64$: Projected dimension.
- **Key Insight:** These fixed orthogonal matrices preserve relative differences and avoid the need for backpropagation.

Attention Calculation & Feature Update

Attention Calculation: For node f_i , the attention weight for neighbor f_j in the dynamic graph is:

$$\text{Atten}_{ij} = \text{Softmax} \left(\frac{(f_i Q)(f_j K)^\top}{\sqrt{d_a}} \right) (f_j^{(t)} V)$$

Temporal Connectivity Constraint

An indicator function $E(i, j)$ ensures attention respects the event-induced topology. This constraint prevents **attention dispersion** to irrelevant frames.

Iterative Feature Update: Features are updated iteratively using the computed attention and connectivity:

$$f_i^{(t+1)} = f_i^{(t)} + \sum_{j \in \mathcal{N}_i} \text{Atten}_{ij} \cdot E(i, j)$$

Feature Centering

Feature Centering: After each iteration, features are centered to stabilize the representations across the video frames F :

$$f_i^{(t+1)} \leftarrow f_i^{(t+1)} - \frac{1}{|F|} \sum_{k \in F} f_k^{(t+1)}$$

Summary of Propagation

Through iterative graph attention updates, the feature propagation maintains:

- **Global divergence**
- **Local consistency**

This balance strictly preserves the relative differences that are **critical for anomaly detection**.

3) Statistical Boundary Detection

Objective

To identify event transitions via **temporal discontinuity analysis** using the enhanced graph attention network features.

Composite Divergence Metric:

For consecutive frames $(i, i + 1)$, a composite divergence metric s_i is computed:

$$s_i = \underbrace{\|f_{i+1} - f_i\|_2^2}_{\text{Feature Magnitude Jump}} + \underbrace{\left(1 - \frac{f_i^\top f_{i+1}}{\|f_i\| \|f_{i+1}\|}\right)}_{\text{Cosine Dissimilarity}}$$

- **Squared L2 Term:** Amplifies abrupt feature-space jumps (amplitude).
- **Cosine Term:** Detects directional changes in the manifold.
- **Result:** A dual strategy that robustly captures both amplitude and directional discontinuities caused by anomalies.

Noise Suppression & Signal Ratio

Challenge

The raw dissimilarity score s_i contains high-frequency noise from camera shake or minor motions.

1. Savitzky-Golay Filter (Smoothing):

Applied with a window width $w = 60$ to suppress noise while preserving true boundaries:

$$\tilde{s}_i = \frac{1}{w} \sum_{k=-w/2}^{w/2} a_k s_{i+k}$$

* a_k derives from a quadratic polynomial fit.

2. Moving Average & Signal Ratio:

A moving average μ_i and the final signal ratio r_i are computed to enhance noise robustness:

$$\mu_i = \frac{1}{w} \sum_{k=i-\lfloor w/2 \rfloor}^{i+\lfloor w/2 \rfloor} \tilde{s}_k \quad \Longrightarrow \quad r_i = \frac{\tilde{s}_i}{\mu_i}$$

Adaptive Thresholding & Boundary Extraction

Mitigating Outlier Sensitivity

To handle occlusions or jitter, **Median Absolute Deviation (MAD)** is used for adaptive thresholding instead of standard variance.

Threshold Calculation:

$$\text{MAD}(r_i) = \text{median}(|r_i - \text{median}(r_i)|)$$

$$M = \text{median}(r_i) + k \times \text{MAD}(r_i)$$

- **Parameter k :** Set to 3, aligning with the **3σ principle**, which captures 99.7% of normal variation.
- **Decision Rule:** Event boundaries are identified when $r_i > M$.
- **Post-processing:** Adjacent boundary points are merged to prevent over-segmentation and refine accuracy.

4) Event-Centric Anomaly Scoring

Limitations of Existing Methods

Current methods rely on discrete frame-level analysis or global video processing using **non-adaptive segmentation**.

- **Overly short fragments:** Lose crucial contextual information.
- **Excessively long units:** Weaken spatial feature representation.
- **Impact:** Hinders the ability of Visual Language Models (VLMs) to interpret videos effectively.

Proposed Solution: Event Semantic Units

- Constructs event semantic units as **spatiotemporal primitives**.
- Creates an optimized feature representation framework specifically tailored for visual-language models.

Hierarchical Prompting Framework

Semantic-Driven Hierarchical Prompting

Directs Multimodal Large Language Models (MLLMs) to produce structured outputs for event unit analysis.

Two-Stage Reasoning Process:

1. **Content Description:** Identifies surface and latent semantic features to generate descriptive text of the video content.
2. **Anomaly Scoring:** Derives an anomaly score based on the description, enabling cross-modal evaluation against predefined criteria.

Key Advantages:

- Establishes a **self-correction mechanism** through two-stage reasoning.
- Ensures systematic score derivation based on video content analysis.
- Maintains contextual relevance and significantly reduces scoring inaccuracies.

EventVAD: Overall Pipeline

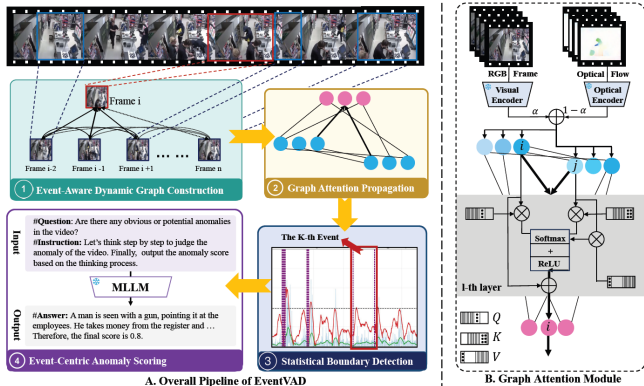


Figure: EventVAD comprises four core modules: (1) Dynamic Graph Construction for establishing spatiotemporal correlations, (2) Graph Attention Propagation for refining node features, (3) Statistical Boundary Detection for identifying event transitions and boundaries, and (4) Event-Centric Scoring for evaluating anomalies via MLLMs within semantic event units.

Implementation Details

Setting Up

We adopt the CLIP ViT-B/16 model to extract 512-dimensional semantic embeddings, and the pre-trained RAFT model to obtain 128-dimensional optical flow features, with $\text{FPS} = 30$. We also use bilinear interpolation for spatial alignment to maintain resolution consistency.

Backbone and Hyperparameters

- **Backbone:** VideoLLaMA2.1-7B-16F
- **Semantic-motion fusion coefficient:** $\alpha = 0.75$
- **Time decay factor:** $\gamma = 0.6$
- **Graph attention propagation iterations:** single iteration
- **Hardware:** single NVIDIA A800 (80GB) GPU

Results on UCF-Crime

Method	Backbone	Supervised	AUC (%)
RTFM [48]	I3D-RGB	Weakly Supervised	84.03
MSL [20]	VideoSwin-RGB	Weakly Supervised	85.62
S3R [57]	I3D-RGB	Weakly Supervised	85.99
MGFN [4]	VideoSwin-RGB	Weakly Supervised	86.98
SSRL [16]	I3D-RGB	Weakly Supervised	87.43
CLIP-TSA [14]	ViT	Weakly Supervised	87.58
BODS [54]	I3D-RGB	One Class	68.26
GODS [54]	I3D-RGB	One Class	70.46
Tur [50]	ResNet	Unsupervised	65.22
Tur [51]	ResNet	Unsupervised	66.85
DyAnNet [47]	I3D	Unsupervised	79.76
ZS CLIP [33]	ViT	Training Free	53.16
ZS ImageBind (Image) [7]	ViT	Training Free	53.65
ZS ImageBind (Video) [7]	ViT	Training Free	55.78
LLaVA-1.5 [23]	ViT	Training Free	72.84
Video-Llama2 [74]	ViT	Training Free	74.42
LAVAD [67]	ViT	Training Free	78.33
EventVAD	ViT	Training Free	82.03

Results on XD-Violence

Method	Backbone	Supervised	AP (%)
Wu et al. [59]	I3D-RGB	Weakly Supervised	73.20
Wu and Liu [58]	I3D-RGB	Weakly Supervised	75.90
RTFM [48]	I3D-RGB	Weakly Supervised	77.81
MSL [20]	VideoSwin-RGB	Weakly Supervised	78.58
S3R [57]	I3D-RGB	Weakly Supervised	80.26
MGFN [4]	VideoSwin-RGB	Weakly Supervised	80.11
Hasan et al. [9]	AERGB	One Class	50.32
Lu et al. [30]	Dictionary	One Class	53.56
BODS [54]	I3D-RGB	One Class	57.32
GODS [54]	I3D-RGB	One Class	61.56
RareAnom [46]	I3D-RGB	Unsupervised	68.33
Blip2 [17]	ViT	Training Free	10.89
ZS CLIP [33]	ViT	Training Free	17.83
ZS ImageBind (Image) [7]	ViT	Training Free	27.25
ZS ImageBind (Video) [7]	ViT	Training Free	25.36
LLaVA-1.5 [23]	ViT	Training Free	50.26
Video-Llama2 [74]	ViT	Training Free	53.57
LAVAD [67]	ViT	Training Free	60.02
EventVAD	ViT	Training Free	64.04

Ablation Study of Hyperparameters

$\gamma \backslash \alpha$	0	0.25	0.5	0.75	1
0.0	77.92	77.13	78.47	78.91	77.34
0.2	78.22	78.85	79.48	80.56	77.93
0.4	78.95	79.12	79.95	81.78	79.46
0.6	79.89	80.67	81.57	82.03	79.69
0.8	78.92	79.54	80.38	81.34	79.62
1.0	77.12	78.81	78.75	80.97	78.97

Observation

The best performance is achieved at $\alpha = 0.75$ and $\gamma = 0.6$, indicating that balanced semantic-motion fusion and moderate temporal decay are crucial for robust anomaly detection.